Numerical processing of financial data – Session 6

13/02/2026

- Recall the theta-gamma approx. of PnL with delta-hedging:

$$\widetilde{PnL}_{t_i,t_{i+1}} = \frac{1}{(1+r\Delta t)^{i+1}} \frac{1}{2} \partial_{ss} P_\sigma(t_i, S_{t_i}) S_{t_i}^2 \left(\sigma^2 \Delta t - \left(\frac{\Delta S_i}{S_{t_i}}\right)^2\right)$$

where $\boxed{P_\sigma(t,s)}$ is the Black-Scholes price function with parameter σ we use to price and delta-hedge the option

realized asset return

- So far, we have chosen:

$$\sigma^2 \Delta t = \text{average of squared returns of } S \text{ over } \Delta t$$

- Remark: now that we have the Black-Scholes PDE and pricing functions, we are free to choose as pricing function the solution

$$P_{\hat{\sigma}}(t,s)$$

to the BS PDE with some other volatility parameter $\hat{\sigma}$.

- Q: what is the consequence of such a choice?

▷ <u>PnL analysis</u>:

$$\mathrm{PnL}_{t_i,t_{i+1}} = \left(-P_{\hat\sigma}(t_{i+1}, S_{t_{i+1}}) + V_{t_{i+1}}\right) - \left(-P_{\hat\sigma}(t_i, S_{t_i}) + V_{t_i}\right)$$

$$= -\Delta P_i + \Delta V_i$$

same steps: Taylor development for $P_{\hat\sigma}(\cdot,\cdot)$
Use the dynamics of self-financing portfolio

Get:

$$\widetilde{\mathrm{PnL}}_{t_i,t_{i+1}} = \frac{1}{(1+r\Delta t)^{i+1}}\left[-\Theta(t_i, S_{t_i})\Delta t - \frac{1}{2}\partial_{ss} P_{\hat\sigma}(t_i, S_{t_i})(\Delta S_i)^2\right]$$

where, as usual,

$$\Theta(t,s) = \partial_t P_{\hat\sigma} + rs\,\partial_s P_{\hat\sigma} - rP_{\hat\sigma}$$

<u>Now</u>, because of the PDE_{BS} with vol parameter $\hat\sigma$,

$$-\Theta(t,s) = \frac{1}{2}\partial_{ss} P_{\hat\sigma}(t,s)\cdot \hat\sigma^2 s^2$$

the parameter we use to price and hedge

so that

$$PnL_{t_i,t_{i+1}} \overset{\circledast}{=} \frac{1}{(1+r\Delta t)^{i+1}} \frac{1}{2} \partial_{ss} P_{\hat{\sigma}}(t_i, S_{t_i}) S_{t_i}^2 \left(\hat{\sigma}^2 \Delta t - \left(\frac{\Delta S_i}{S_{t_i}} \right)^2 \right)$$

↳ same expression as before, now with $\hat{\sigma}$ instead of σ.

- Note that the expression $\circledast$ provides an approx. of the PnL <u>even if we do not make the</u> assumption of constant realized vol for S:

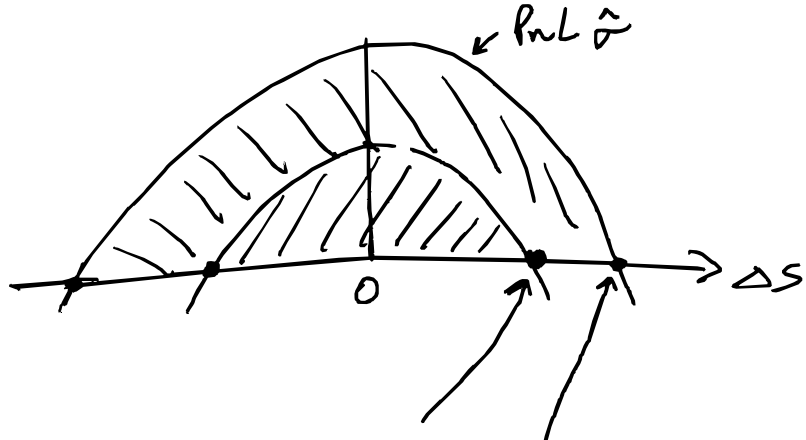
$$E[(R_{t,t+\Delta t})^2 \mid F_t] = \sigma^2 \Delta t \quad , \forall t$$

<u>Consequence</u>: the theta-gamma approx of the PnL $\circledast$ holds whatever the dynamics of the asset S in the market (it is "<u>model-free</u>")

▷ The use of the pricing function $P_{\hat{\sigma}}(t,s)$ leads to <u>new break-even points</u>:

$$|\Delta S_i^*| = S_{t_i}\, \hat{\sigma}\sqrt{\Delta t} \qquad (\hat{\sigma} \text{ instead of } \sigma)$$

▷ This means that, if $\partial_{ss} P_{\hat{\sigma}}(t,S) \geq 0$, then taking $\boxed{\hat{\sigma} > \sigma}$ we push the PnL towards positive values:



old BE points : $|\Delta S_i^*| = \sigma\sqrt{\Delta t}\, S_i$

new BE points: $|\Delta S_i^*| = \hat{\sigma}\sqrt{\Delta t}\, S_{t_i}$

▷ <u>Overall</u> : $\hat{\sigma}$ used in the pricing PDE is a parameter that allows to offset the theta-gamma PnL and therefore to control the impact of $(\Delta S)^2$ on our position

▷ Note that it can be shown that

$$\hat{\sigma} > \sigma$$

implies

$$\boxed{P_{\hat{\sigma}}(t,s) > P_{\sigma}(t,s)}$$

under the condition $\partial_{ss} P_{\sigma'}(t,s) \geq 0$, $\forall t, \forall s, \forall \sigma'$

↳ with $\hat{\sigma} > \sigma$, we sell the option for a larger price (more difficult)

▷ <u>For the expectation of the PnL</u>, we have

$$E\left[\widetilde{PnL}_{t_i,t_{i+1}} \,\middle|\, \mathcal{F}_{t_i}\right]$$

$$= \frac{1}{(1+r\Delta t)^{i+1}} \frac{1}{2} \partial_{ss} P_{\hat{\sigma}}(t_i, S_{t_i}) (S_{t_i})^2 \left(\hat{\sigma}^2 \Delta t - E\left[\left(\frac{\Delta S_i}{S_i}\right)^2 \middle| \mathcal{F}_{t_i}\right]\right)$$

if we now make, once again, the assumption on the realized variance:

$$E\left[\left(\frac{\Delta S_i}{S_{t_i}}\right)^2 \middle| \mathcal{F}_{t_i}\right] = \sigma^2 \Delta t \quad , \forall t_i$$

we get

$$E\left[\widetilde{PnL}_{t_i,t_{i+1}} \middle| \mathcal{F}_{t_i}\right] = \frac{1}{(1+r\Delta t)^{i+1}} \frac{1}{2} \partial_{ss} P_{\hat{\sigma}}(t_i, S_{t_i}) (S_{t_i})^2 (\hat{\sigma}^2 - \sigma^2) \Delta t$$

Hence, we have:

$$\boxed{E\left[PnL_{t_i,t_{i+1}} \middle| \mathcal{F}_i\right] > 0} \quad \text{if } \hat{\sigma} > \sigma$$

instead of $E\left[\text{———} \middle| \mathcal{F}_i\right] = 0$ if $\hat{\sigma} = \sigma$

more difficult to realize in practice